

Questions and answers ANN

Choose the suitable choice for the following statements

1. The areas in which neural networks are currently being applied are: -----

a. speech Recognition b. searching c. control problems ☒ d. both a&c

2. In neuron input paths called -----

☒ a. dendrites b. synapses c. weights d. axon

3. ----- is the junction between the (axon) of the neuron and the dendrites of the other neuron

☒ a. synapse b. dendrites c. weight d. activation function

4. Information processing occurs at many simple elements called -----

a. weight b. path ☒ c. neuron d. axon

5. Each neuron applies an action -----

6. A Neural network is characterized by: -----

a. Training Learning Algorithm b. Activation function c. architecture ☒ d.

all a,b,c

7. One of ANN properties is

a. serial ☒ b. Fault tolerance c. activation function d. supervision

8. There are ---- types of learning in which the weights organize themselves according to the task to be learnt, these types are-----, -----, -----

a. 3, supervised, unsupervised, activated b. 3, self learned, activated, supervised c. 3, unsupervised, self learned, activated ☒ d. 3, supervised, unsupervised, self learned

9. In the unsupervised learning, the correct final vector is not -----, but instead the ----- are changed through random numbers

a. learned, weight ☒ b. specified, weights c. activated, weights
d. changed, weights

10. In the supervised learning, at every step the system is informed about the -----
- output vector. The ----- are changed according to a formula

- ☒ a. exact, weights b. next, weights c. only, weights d. nearly, weights

11. Neural nets are often classified as

- a. single layer, double layer b. single layer, triple layer ☒ c. single layer, multi layer d. single layer, all layer

12. In determining the number of layers, the input units are ----- as a layer

- ☒ a. not specified ☒ b. not counted c. not determined d. not countered

13. ----- units are between the input units and the output units in the -----
---- net.

- ☒ a. hidden, multi layer b. hidden, double layer c. hidden, triple layer, d.
hidden, single layer

14. the hard limiter activation function is -----& -----

- a. binary, tri-polar bi-polar, tri-polar ☒ c. binary, bi-polar d. binary, single polar

15. the input to binary hard limits is -----& ----- while in the bipolar is -----& -----

- ☒ a. 1, 0, 1,-1 b. 0, 1, 1/2,-1/2 c. 0,1,2,-2 d. 1,-1,0,1

Example2 – Delta learning rule

Perform two training steps of neural network, using the delta learning rule for $\lambda = 1$ and $c = 0.25$. Train the network using the following data pairs

$$\left(\mathbf{x}_1 = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, d_1 = -1 \right), \left(\mathbf{x}_2 = \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}, d_2 = 1 \right)$$

The initial weights are $\mathbf{W}_1 = [1 \ 0 \ 1]^t$ and $f(\text{net})$ is bipolar continuous activation function.

Solution

$F(\text{net}) = \text{bipolar activation function} =$

$$\frac{2}{1 + \exp(-\lambda x)} - 1$$

$$F'(\text{net}) = \frac{1}{2}[1 - (O_i)^2]$$

$$\text{Net}_1 = \mathbf{X}_1 * \mathbf{W}_1 = [2 \ 0 \ -1] * [1 \ 0 \ 1] = 1$$

$$O_1 = f(\text{net}_1) = 0.47$$

Since $\text{Sgn}(O_1)$ not equal d_1 (-1) the correction is necessary

$$F'(\text{net}_1) = \frac{1}{2}(1 - O^2) = \frac{1}{2}(1 - (0.47)^2) = 0.39$$

$$\mathbf{W}_2 = \mathbf{W}_1 + c(d_i - O_i) F'(\text{net}_1) * \mathbf{X}_1$$

$$\begin{aligned} \mathbf{W}_2 &= [1 \ 0 \ 1] + 0.25(-1 - 0.47) * 0.39 * [2 \ 0 \ -1] = [1 \ 0 \ 1] + (-0.14) * [2 \ 0 \ -1] \\ &= [1 \ 0 \ 1] + [-0.28 \ 0 \ 0.14] \quad \mathbf{W}_2 = [0.72 \ 0 \ 1.14] \end{aligned}$$

$$\text{Net}_2 = \mathbf{X}_2 * \mathbf{W}_2 = [1 \ -2 \ -1] * [0.72 \ 0 \ 1.14] = -0.42$$

$$O_2 = f(\text{net}_2) = -0.2$$

Since $\text{Sgn}(O_2)$ not equal d_2 (1) the correction is necessary

$$F'(\text{net}_2) = \frac{1}{2}(1 - O^2) = \frac{1}{2}(1 - (-0.2)^2) = 0.48$$

$$\mathbf{W}_3 = \mathbf{W}_2 + c(d_2 - O_2) F'(\text{net}_2) * \mathbf{X}_2$$

$$\mathbf{W}_3 = [0.72 \ 0 \ 1.14] + 0.25(1 + 0.2) * 0.48 * [1 \ -2 \ -1]$$

$$\mathbf{W}_3 = [0.72 \ 0 \ 1.14] + [1.44 \ -2.88 \ -1.44]$$

$$\mathbf{W}_3 = [2.16 \ -2.88 \ -0.3]$$

Genetic algorithms

1. GA works on two types of spaces -----& -----

Genotype& phenotype

2. The----- is the link between chromosomes and the performance decoded solutions

Selection

3. GA provides ----- search in complex landscapes

Random

4. There are two important issues with respect to search strategies-----
& -----

Exploration& Exploitation

5. Infeasibility refers to the phenomenon that a solution decoded from chromosome lies----- the feasible region of given problem

outside

6. Genetic operators -----& ----- work on genotype space

Crossover& Mutation

7. Genetic operators -----& ----- work on phenotype space

Evolution& Selection

8. To cross the Hamming cliffs in ----- all bits have to be -----

Binary encoding, changed simultaneously

9. In ----- the structure of genotype space is ----- to that in phenotype.

Real number encoding, Identical

10. Integer or literal permutation encoding is best for -----

Combinational optimization

What are the important issues with respect to search strategies in Genetic Algorithms?

- There are two important issues with respect to search strategies:: exploration (investigate new and unknown areas in search space) and exploitation (make use of knowledge of solutions previously found in search space to help in find better solutions).

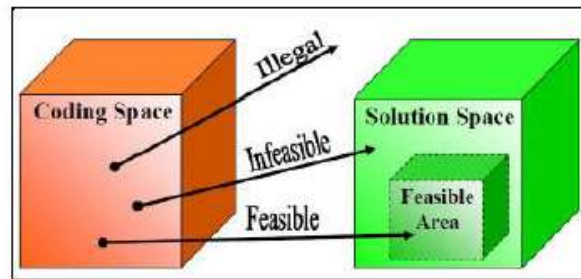
What are Genetic Algorithms? Where GAs work?

- Genetic Algorithms (or simply GAs) are powerful and widely applicable stochastic search and optimization methods based on the concepts of natural selection and natural evaluation.

GAs work on a population of individuals represents candidate solutions to the optimization problem. These individual are consists of a strings (called chromosomes) of genes.

Explain with diagram infeasibility and illegality?

- **Infeasibility** refers to the phenomenon that a solution decoded from chromosome lies outside the feasible region of given problem. Penalty methods can be used handle infeasible chromosomes. One of these methods is by force genetic algorithms to approach optimal form both sides of feasible and infeasible regions.
- **Illegality** refers to the phenomenon that a chromosome does not represent a solution to a given problem. Repair techniques are usually adopted to convert an illegal chromosome to legal one.



Tutorial Sheet of Genetic Algorithm

Part A: Crossover and Mutation

Q₁: Apply single crossover of 6th point and mutation of 3rd point on the following population :

Parent	Pattern	X	2X-1
P1	00000110		
P2	00000010		
P3	00001000		
P4	00001001		
P5	00000110		
P6	00001111		

Solution:

1. Calculate the both value of **X** and **2X-1** from the above Pattern

Parent	Pattern	X	2X-1
P1	00000110	6	11
P2	00000010	2	3
P3	00001000	8	15
P4	00001001	9	17
P5	00000110	6	11
P6	00001111	15	29

2. After crossover 6th point

Children	Pattern	X	2X-1
Child 1	00000010	2	3
Child 2	00000110	6	11
Child 3	00001001	9	17
Child 4	00001000	8	15
Child 5	00000111	7	13
Child 6	00001110	14	27

3. Apply Mutation 3rd point

Children	Pattern	X	2X-1
Child 1	00000010	2	3
Child 2	00000110	6	11
Child 3	00001001	9	17
Child 4	00001000	8	15
Child 5	00000111	7	13
Child 6	00001110	14	27

4. After Mutation 3rd point

Children	Pattern	X	2X-1
Child 1	00100010	34	67
Child 2	00100110	38	75
Child 3	00101001	41	81
Child 4	00101000	40	79
Child 5	00100111	39	77
Child 6	00101110	46	91

Q₂: For the population below with the fitness function $Y = A+B+C+D-30$, apply the following:

1. Single crossover of 4th point
2. Mutation of 4th and 1st points
3. Repeat No 2 with Mutation of 3rd point

Parent	Pattern		
P1	[12;05;07,15]		
P2	[01;08;20,26]		
P3	[05;10;13,20]		
P4	[07;13;09,11]		

Solution:

Calculate the value of **Y**(Evaluation) from the above Pattern

Parent	Pattern	Evaluation $Y = A+B+C+D-30$
P1	[12;05;07,15]	$12+5+7+15-30=9$
P2	[01;08;20,26]	25
P3	[05;10;13,20]	18
P4	[07;13;09,11]	10

1. After crossover 4th point

Children	Pattern	Evaluation $Y = A+B+C+D-30$
Child 1	[12;05;07,26]	20
Child 2	[01;08;20,15]	14
Child 3	[05;10;13,11]	9
Child 4	[07;13;09,20]	19

2. After Mutation of 4th and 1st points

4th in Ch₂

1st in Ch₄

Randomly Ch₂, Ch₄

Children	Pattern	Evaluation $Y = A+B+C+D-30$
Child 1	[12;05;07,26]	20
Child 2	[01;08;20,07]	6
Child 3	[05;10;13,11]	9
Child 4	[15;13;09,20]	27

3. H.W

Part B: Selection: Roulette Wheel Selection

Q₃: Apply roulette wheel selection for the following population:

Parent	Pattern	X	2X-1
P1	00000110		
P2	00000010		
P3	00001000		
P4	00001001		
P5	00000110		
P6	00001111		

Randomly values

1. 0.2
2. 0.01
3. 0.8
4. 0.21
5. 0.11
6. 0.65

Solution:

Calculate the both value of **X** and **2X-1** from the Pattern

Parent	Pattern	X	2X-1
P1	00000110	6	11
P2	00000010	2	3
P3	00001000	8	15
P4	00001001	9	17
P5	00000110	6	11
P6	00001111	15	29

1. Roulette Selection

Parent	Pattern	X	$F_{obj} = 2X-1$	Fitness Evaluation $F = \frac{1}{1+f_{obj}}$	Fitness Probability $P = \frac{F_{part}}{F_{all}} = \frac{Fitness}{Total\ Fitness}$		Randomly values	New Population
P1	00000110	6	11	$1/1+11=0.083$	$0.083/0.5544=0.15$	0.15	0.2	P2
P2	00000010	2	3	$1/1+3=0.25$	$0.25/0.5544=0.45$	$0.15+0.45=0.6$	0.01	P1
P3	00001000	8	15	0.0625	0.11	0.71	0.8	P4
P4	00001001	9	17	0.055	0.09	0.8	0.21	P2
P5	00000110	6	11	0.0714	0.13	0.93	0.11	P1
P6	00001111	1	29	0.0333	0.06	$0.98=1$	0.65	P3
				Total Fitness =0.5552				

Hint:

1. Objective function= $f(x) = F_{obj}$
2. Fitness of each individual = $Fitness[] = (1 / (1+F_{obj}[]))$
3. Fitness probability $P[i] = Fitness[] / Total$

Q4: For the following population below, apply the following:

1. Roulette wheel selection
2. Single Crossover over 5th point
3. Mutation of 6th point
4. Repeat No(3) to make **Mutations** of 1st point on **Fourth** pattern and 3rd point on **Five** pattern

Randomly values

1. 0.2
2. 0.01
3. 0.8
4. 0.21
5. 0.11
6. 0.65

Solution:

Calculate the both value of **X** and **2X-1** from the above Pattern

Parent	Pattern	X	2X-1
P1	00000110	6	11
P2	00000010	2	3
P3	00001000	8	15
P4	00001001	9	17
P5	00000110	6	11
P6	00001111	15	29

1. Roulette Selection

Parent	Pattern	X	F_obj= 2X-1	Fitness[]= $\frac{1}{1+F_{obj}}$	Fitness Probability p[]= $\frac{Part}{All}$		Randomly values	New Population
P1	00000110	6	11	$1/1+11=0.083$	$0.083/0.5544=0.15$	0.15	0.2	P2
P2	00000010	2	3	$1/1+3=0.25$	$0.25/0.5544=0.45$	$0.15+0.45=0.6$	0.01	P1
P3	00001000	8	15	0.0625	0.11	0.71	0.8	P4
P4	00001001	9	17	0.055	0.09	0.8	0.21	P2
P5	00000110	6	11	0.0714	0.13	0.93	0.11	P1
P6	00001111	15	29	0.0333	0.06	0.98=1	0.65	P3
				Total=0.5552				

Hint:

1. Objective function= $f(x) = F_{obj}$
2. Fitness of each individual = $Fitness[] = (1 / (1+F_{obj}[]))$
3. Fitness probability $P[i] = Fitness[] / Total$

2. Single Crossover over 5th point

New Population

Parent	Pattern	X	2X-1
P1	00000010	2	3
P2	00000110	6	11
P3	00001001	9	17
P4	00000010	2	3
P5	00000110	6	11
P6	00001000	8	15

After Crossover over 5th point

Children	Pattern	X	2X-1
Child 1	00000110	6	11
Child 2	00000010	2	3
Child 3	00000010	2	3
Child 4	00001010	10	19
Child 5	00001000	8	15
Child 6	00000110	6	11

3. After Mutation of 6th point

Children	Pattern	X	2X-1
Child 1	00000010	2	11
Child 2	00000110	6	3
Child 3	00000110	6	3
Child 4	00001110	14	19
Child 5	00001100	12	15
Child 6	00000010	2	11

4. H.W

Q₅: For the following population below with the function $Y = A+B+C+D-30$, apply the following:

1. Roulette wheel selection
2. Single Crossover over 4th point
3. Mutation for 4th and 1st point

Randomly values

1. 0.1
2. 0.7
3. 0.3
4. 0.5

Parent	Pattern		
P1	[12;05;07,15]		
P2	[01;08;20,26]		
P3	[05;10;13,20]		
P4	[07;13;09,11]		

Solution: Calculate the value of **Y**(Evaluation) from the above Pattern

Parent	Pattern	Evaluation $Y = A+B+C+D-30$
P1	[12;05;07,15]	$12+5+7+15-30=9$
P2	[01;08;20,26]	25
P3	[05;10;13,20]	18
P4	[07;13;09,11]	10

1. Roulette Selection

Parent	Pattern	$F_{obj} = Y = A+B+C+D-30$	$Fitness[] = \frac{1}{1+F_{obj}}$	Fitness Probability $p[] = \frac{F_{part}}{F_{all}}$		Randomly values	New population
P1	[12;05;07,15]	9	0.1	0.355	0.355	0.1	P1
P2	[01;08;20,26]	25	0.038	0.135	0.49	0.7	P4
P3	[05;10;13,20]	18	0.053	0.188	0.678	0.3	P1
P4	[07;13;09,11]	10	0.09	0.32	1	0.5	P3
			Total Fitness = 0.2819				

The new population will be:

Parent	Pattern
P1	[12;05;07,15]
P2	[07;13;09,11]
P3	[12;05;07,15]
P4	[05;10;13,20]

Hint:

1. Objective function= $f(x) = F_{obj}$
2. Fitness of each individual = $Fitness[] = (1 / (1+F_{obj}[]))$
3. Fitness probability $P[i] = Fitness[] / Total$

2. After Single Crossover 4th point

Children	Pattern
Child 1	[12;05;07,11]
Child 2	[07;13;09,15]
Child 3	[12;05;07, 20]
Child 4	[05;10;13,15]

3. Mutation for 4th and 1st point

By assuming the Mutation Rate= Population size $\times$ rate; rate=0.4 (Given)

$$= 4 \times 0.4 = 2$$

Choose randomly c[2], c[4]

Point 4,1

Children	Pattern
Child 1	[12;05;07,11]
Child 2	[07;13;09,15]
Child 3	[12;05;07, 20]
Child 4	[05;10;13,15]

After Mutation

Children	Pattern
Child 1	[12;05;07,11]
Child 2	[07;13;09,05]
Child 3	[12;05;07, 20]
Child 4	[15;10;13,15]

The new population can be represented as Roulette Selection (H.W)

Parent	Pattern	$F_{obj} = Y = A+B+C+D-30$	$Fitness[] = \frac{1}{1+f_{obj}}$	$Fitness\ Probability\ p[] = \frac{Part}{All}$	Randomly	New population
P1	[12;05;07,11]				0.1	P1
P2	[07;13;09,05]				0.7	P4
P3	[12;05;07,20]				0.3	P1
P4	[15;10;03,15]				0.5	P3
			Total=			

Hint:

1. Objective function= $f(x) = F_{obj}$
2. Fitness of each individual = $Fitness[] = (1 / (1+F_{obj}[]))$
3. Fitness probability $P[i] = Fitness[] / Total$

Q₆:H.W: Suppose a genetic algorithm uses populations as in pattern below with fitness function of individual x: $f(x) = a + 2b + 3c + 4d$

$$x_1 = [a;b;c;d] = [12;05;23;08]$$

$$x_2 = [a;b;c;d] = [02;21;18;03]$$

$$x_3 = [a;b;c;d] = [10;04;13;14]$$

$$x_4 = [a;b;c;d] = [20;01;10;06]$$

1. Calculate the objective function of each individual
2. Compute the fitness of each individual
3. Calculate the fitness probability of each individual
4. Apply Single Crossover of 2nd point
5. Apply Mutation for 1st and 3rd point

Hint:

1. Objective function= $f(x) = F_{obj}$
2. Fitness of each individual = $Fitness[] = (1 / (1+F_{obj}[]))$
3. Fitness probability $P[i] = Fitness[] / Total$

Part C: Questions and Answers

Q₁: What are the important issues with respect to search strategies in Genetic Algorithms?

Answer:

There are two important issues with respect to search strategies:

1. Exploration (investigate new and unknown areas in search space)
2. Exploitation (make use of knowledge of solutions previously found in search space).

Q₂: What are Genetic Algorithms? Where GAs work?

Answer:

Genetic Algorithms (or simply GAs) are powerful and widely applicable stochastic search and optimization methods based on the concepts of natural selection and natural evaluation.

GAs work on a population of individuals represents candidate solutions to the optimization problem.

Q₃: State the Encoding methods? Explain each.

Answer:

Encoding methods can be classified as follows:

- 1) Binary encoding
- 2) Real-number encoding
- 3) Integer or literal permutation encoding

Q₄: What are the two requirements should a problem satisfy in order to be suitable for solving it by a GA?

Answer:

1. The fitness function can be well-defined.
2. Solutions should be decomposable into steps (building blocks) which could be then encoded as chromosomes.

Q₆: What is the general structure of the Genetic Algorithms?

Answer:

The general structure of the Genetic algorithms is as follow:

Begin

{

t=0;

Initialize P(t);

Evaluate P(t);

While (not termination condition) do

Begin

{

Apply crossover and mutation to P(t) to yield C(t);

Evaluate C(t);

Select P(t+1) from P(t) and C(t);

t=t+1;

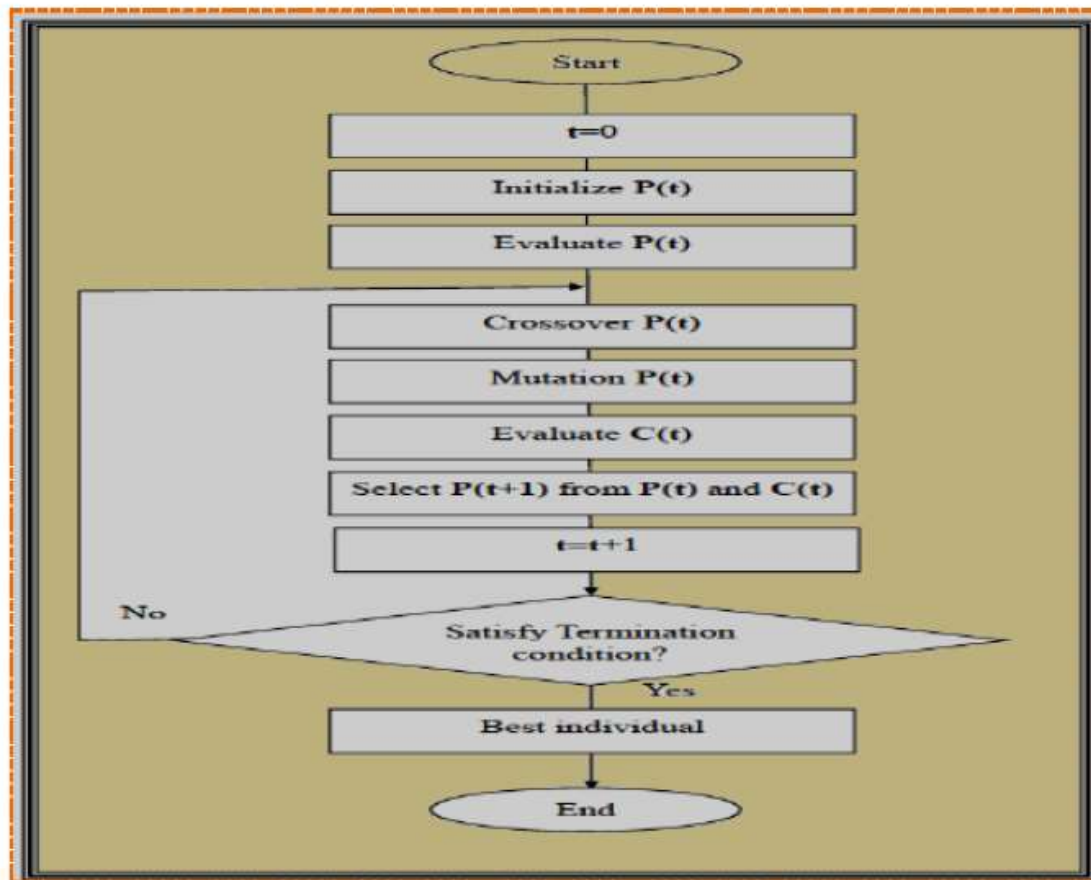
}

End

}

End

Q7. Based on the flowchart explains how genetic algorithms work?



Q₈: List Six Applications of Genetic Algorithms

Answer:

1. Optimization
2. Automatic Programming
3. Machine and robot learning
4. Economic models
5. Immune system models
6. Population genetics models

Q₉: Mention the most Advantages and Limitations of Genetic Algorithm

The advantages of genetic algorithm includes,

1. Parallelism
2. Liability
3. Solution space is wider
4. The fitness landscape is complex
5. Easy to discover global optimum

The limitation of genetic algorithm includes,

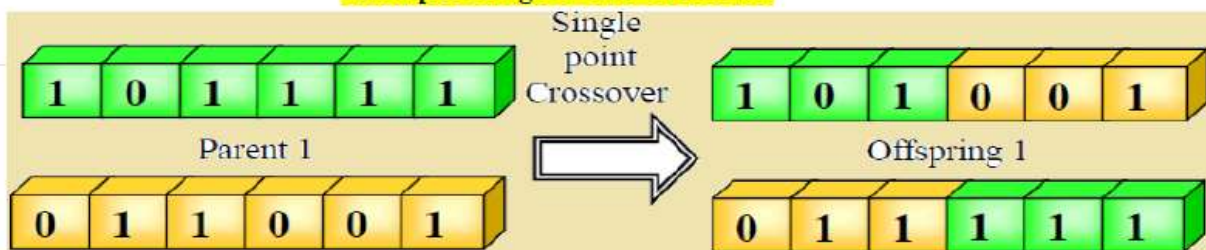
1. The problem of identifying fitness function
2. Definition of representation for the problem
3. Premature convergence occurs
4. The problem of choosing the various parameters like the size of the population, Mutation rate, cross over rate, the selection method and its strength.
5. Cannot use gradients

Q₁₀: State the types of crossover and give **only one** example for any type.

Answer:

1. **Single Point Crossover**
2. **Multi Point Crossover**
3. **Uniform Crossover**

Example: Single Point Crossover

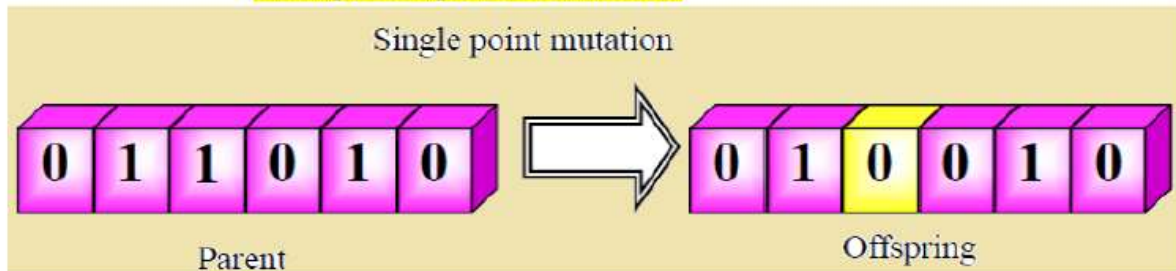


Q₁₁: State the types of mutation and give **only one** example for any type

Answer:

1. **Single Point Mutation**
2. **Multi Point Mutation**

Example: Single Point Mutation



Q₁₂: Choose the suitable words for the following point:

1. GA works on two types of spaces &
2. The is the link between chromosomes and the performance decoded solutions.
3. GA provides search in complex landscapes

4. There are two important issues with respect to search strategies
..... &
5. Infeasibility refers to the phenomenon that a solution decoded from
chromosome lies the feasible region of given problem
6. Genetic operators&work on genotype space
7. To cross the Hamming cliffs in all bits have to be
.....
8. Genetic operators&work on phenotype space
9. In the structure of genotype space is to that in
phenotype
10. Integer or literal permutation encoding is best for